

### Version 0

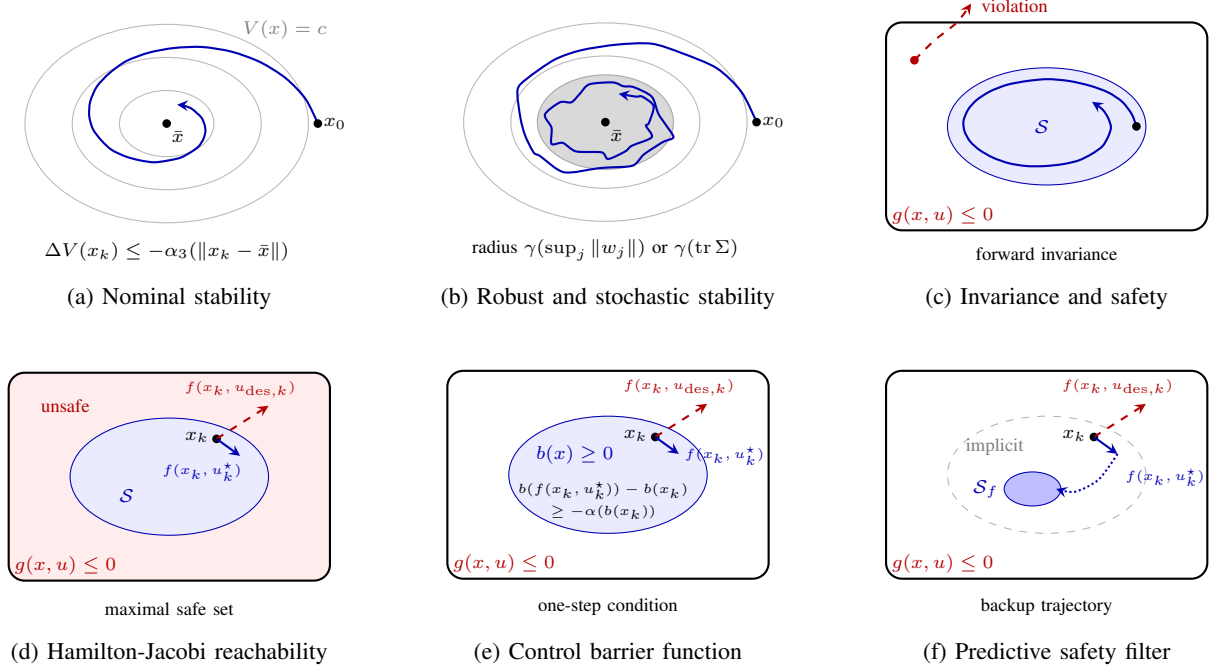


Fig. 1: Illustration of the stability and safety notions of this section. (a) Nominal stability: a Lyapunov function  $V$  decreases along closed-loop trajectories (level sets in gray) and the state converges to the equilibrium  $\bar{x}$ . (b) Robust and stochastic stability: under disturbances  $w$ , the state converges to a neighbourhood of  $\bar{x}$  (shaded), whose size depends on the disturbance bound or the noise covariance. (c) Safety: constraint satisfaction at one time instant does not imply safety (red dashed); safety is guaranteed by an invariant set  $S$  contained in the constraints. (d)–(f) Safety filters: the desired input  $u_{\text{des},k}$  would leave the safe set and is modified to  $u_k^*$ . HJ reachability computes the maximal safe set  $S$  offline and its complement (red) contains the states from which a constraint violation cannot be avoided. A CBF certifies invariance of  $\{b(x) \geq 0\}$  by the one-step condition on  $b(x_{k+1})$ . A PSF does not compute the safe set (dashed) but certifies safety by a backup trajectory (dotted) into a smaller terminal set  $S_f$ .

### Version 1

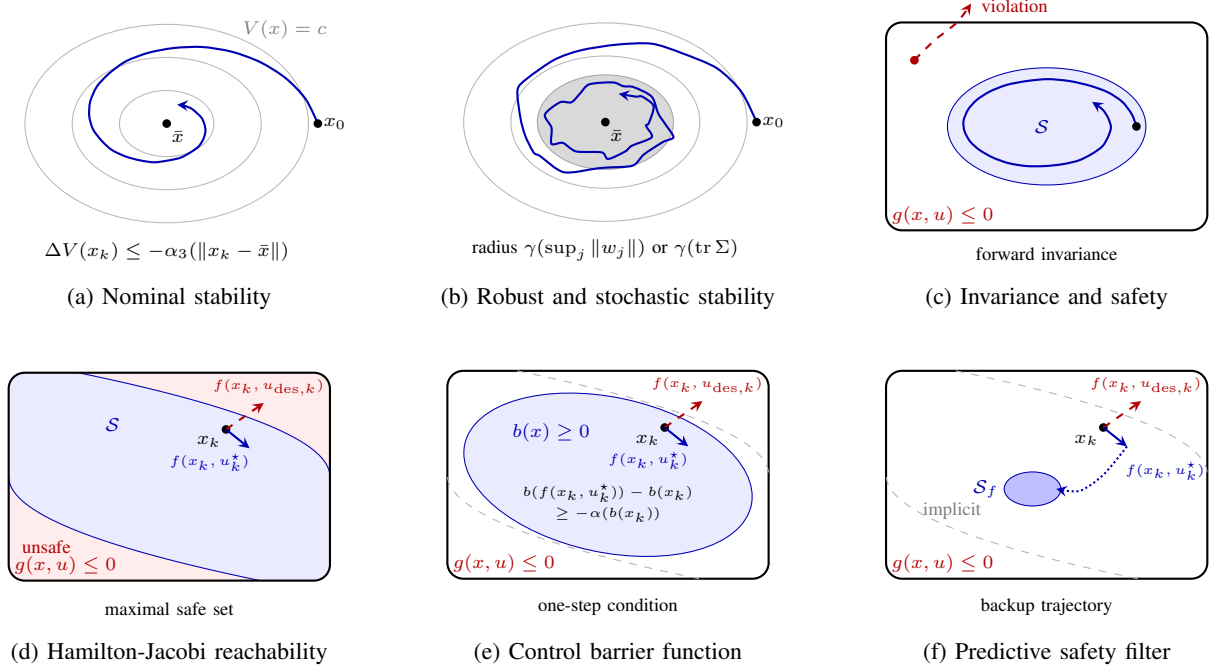


Fig. 2: Illustration of the stability and safety notions of this section. (a) Nominal stability: a Lyapunov function  $V$  decreases along closed-loop trajectories (level sets in gray) and the state converges to the equilibrium  $\bar{x}$ . (b) Robust and stochastic stability: under disturbances  $w$ , the state converges to a neighbourhood of  $\bar{x}$  (shaded), whose size depends on the disturbance bound or the noise covariance. (c) Safety: constraint satisfaction at one time instant does not imply safety (red dashed); safety is guaranteed by an invariant set  $S$  contained in the constraints. (d)–(f) Safety filters: the desired input  $u_{\text{des},k}$  would leave the safe set and is modified to  $u_k^*$ . HJ reachability computes the maximal safe set  $S$  offline and its complement (red) contains the states from which a constraint violation cannot be avoided. A CBF certifies invariance of  $\{b(x) \geq 0\}$  by the one-step condition on  $b(x_{k+1})$ . A PSF does not compute the safe set (dashed) but certifies safety by a backup trajectory (dotted) into a smaller terminal set  $S_f$ .

## Version 2

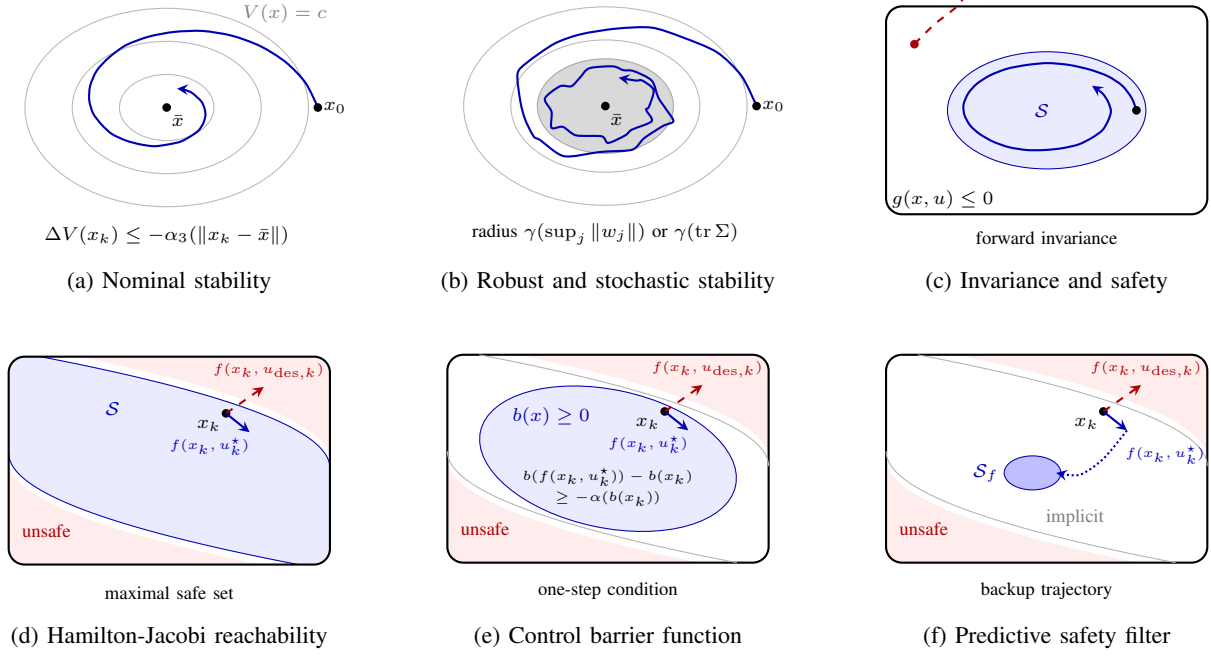


Fig. 3: Illustration of the stability and safety notions of this section. (a) Nominal stability: a Lyapunov function  $V$  decreases along closed-loop trajectories (level sets in gray) and the state converges to the equilibrium  $\bar{x}$ . (b) Robust and stochastic stability: under disturbances  $w$ , the state converges to a neighbourhood of  $\bar{x}$  (shaded), whose size depends on the disturbance bound or the noise covariance. (c) Safety: constraint satisfaction at one time instant does not imply safety (red dashed); safety is guaranteed by an invariant set  $S$  contained in the constraints. (d)–(f) Safety filters: the desired input  $u_{\text{des},k}$  would leave the safe set and is modified to  $u_k^*$ ; the unsafe states are shaded red. HJ reachability computes the maximal safe set  $S$  offline, the largest invariant set within the constraints. A CBF certifies invariance of  $\{b(x) \geq 0\}$  by the one-step condition on  $b(x_{k+1})$ . A PSF does not compute the safe set (gray) but certifies safety by a backup trajectory (dotted) into a smaller terminal set  $S_f$ .